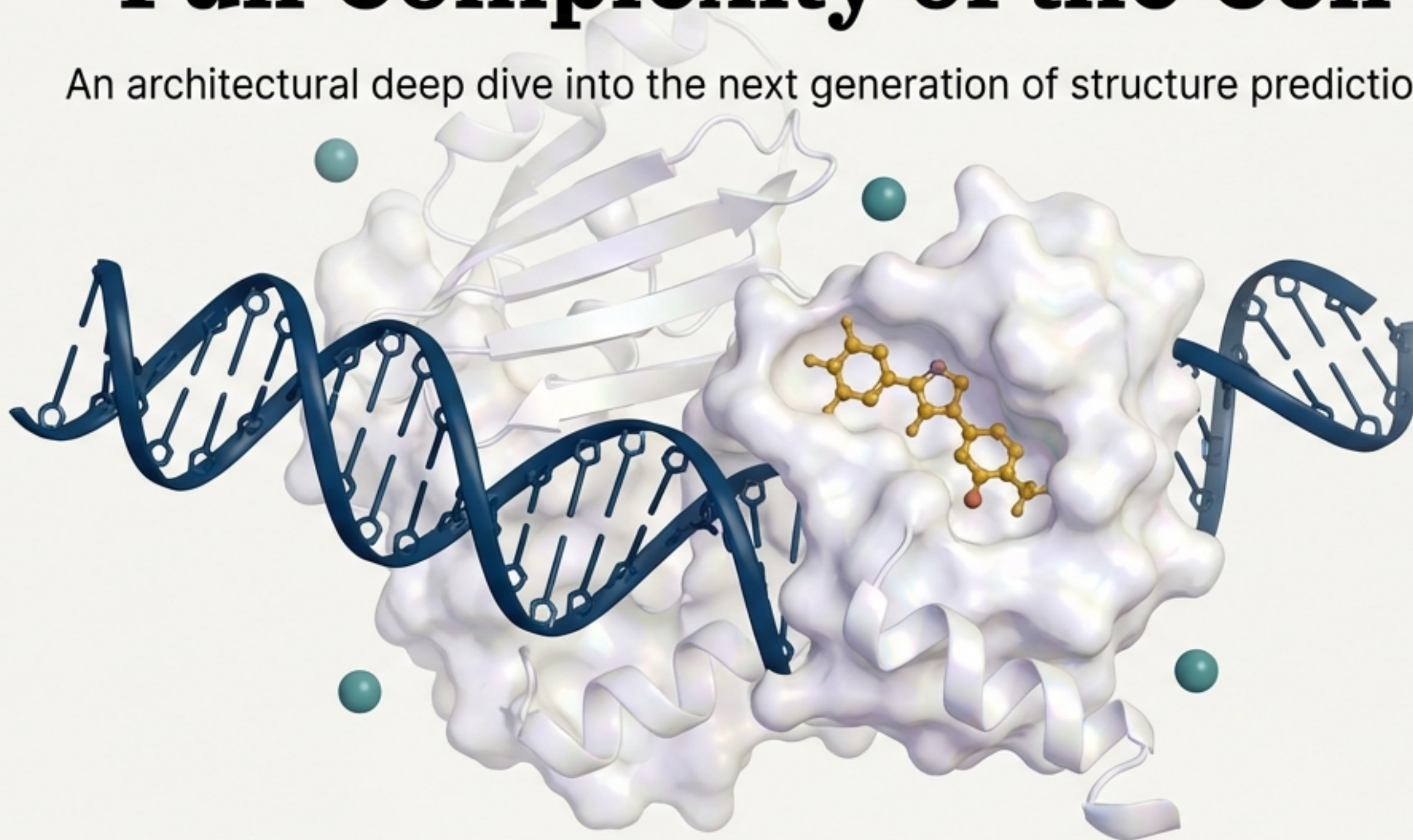


AlphaFold3: Modeling the Full Complexity of the Cell

An architectural deep dive into the next generation of structure prediction.



From Proteins to All of Life's Molecules

AlphaFold2



Proteins

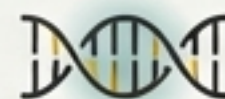


Protein Complexes

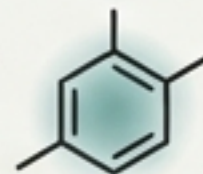
AlphaFold3



Proteins



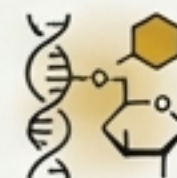
Nucleic Acids (DNA/RNA)



Small Molecules (Ligands)



Ions



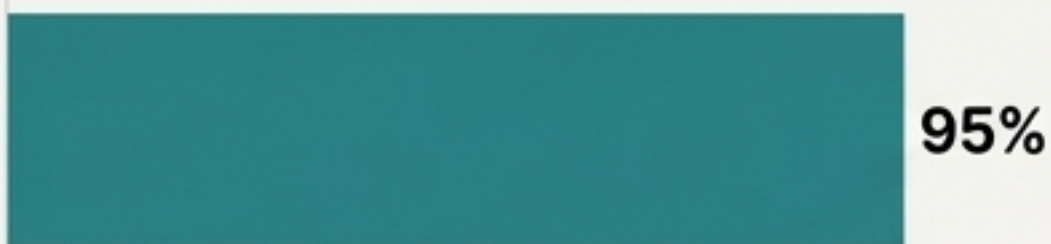
Covalent Modifications

AlphaFold3 introduces a single, unified model that predicts the structure of virtually any biomolecular complex, moving far beyond the protein-centric world of its predecessor.

High Success Rates Across a Diverse Molecular Range

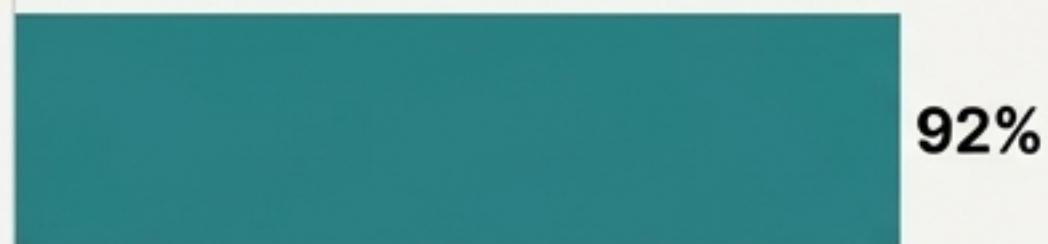
Ligand Binding

(vs. PoseBusters benchmark, 25 predictions)



Nucleic Acid Interactions

(Accuracy > specified threshold)

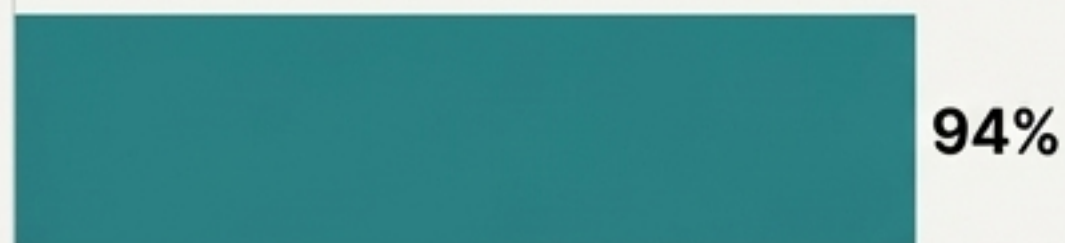


Covalent Modifications

(e.g., glycosylations, phosphorylations)



Protein-Protein Interfaces

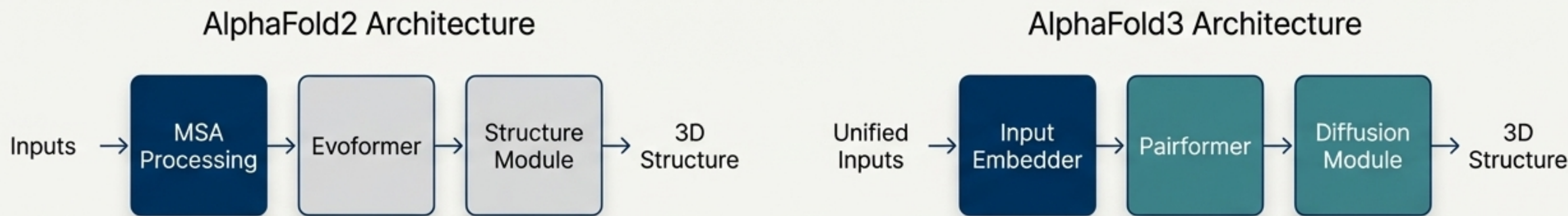


A Note on Sampling

Difficult targets like antibodies required 5,000 predictions ("seeds") to achieve a ~60% success rate.

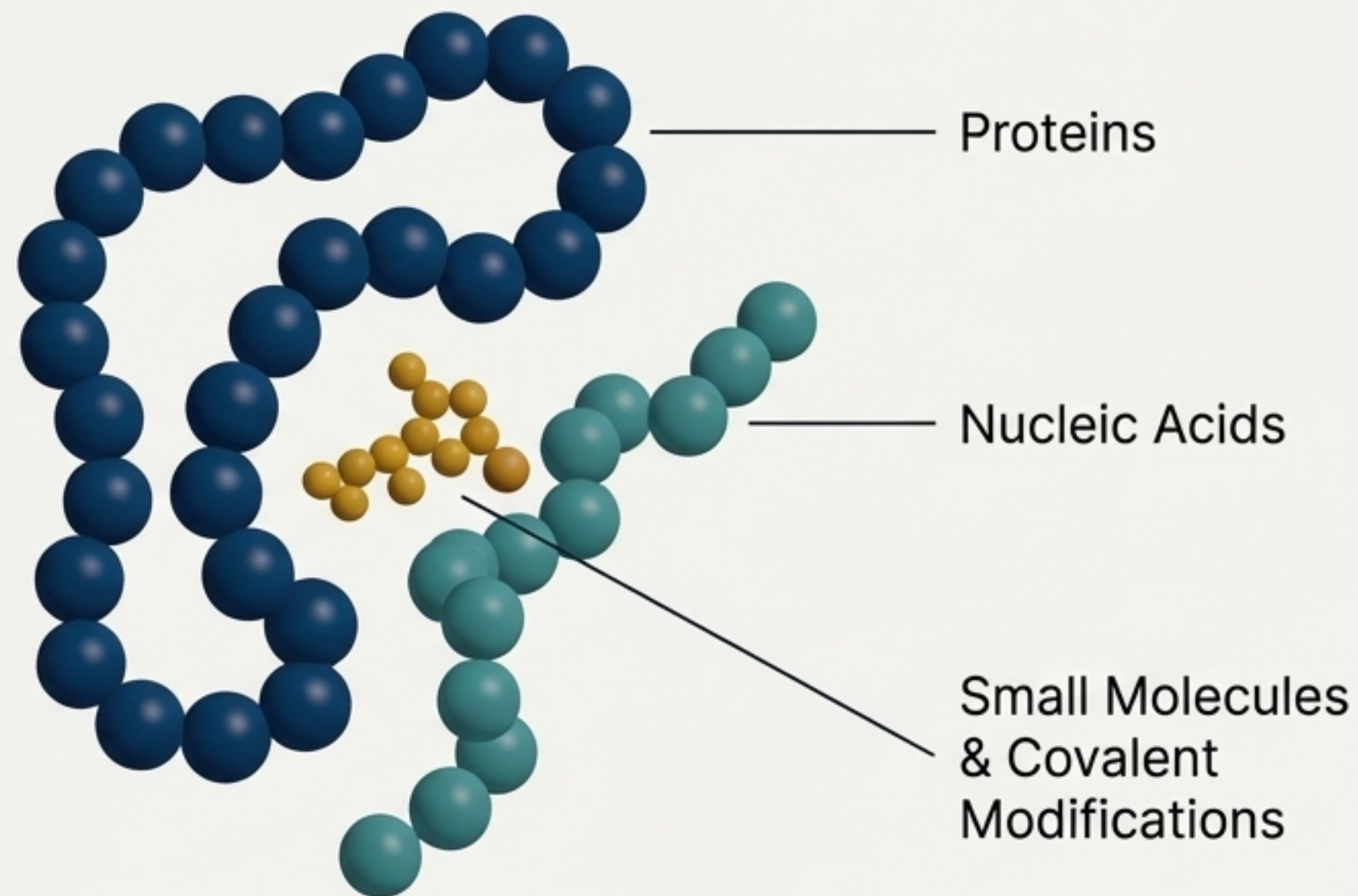
This highlights that achieving high accuracy on challenging systems demands significant computational sampling.

A Redesigned Engine: From Evoformer to Pairformer and Diffusion



AF3 replaces the core of the AF2 architecture. Most critically, it swaps the deterministic Structure Module for a generative Diffusion model, fundamentally changing how structures are produced.

A Universal Language for Molecules: Unified Tokenization



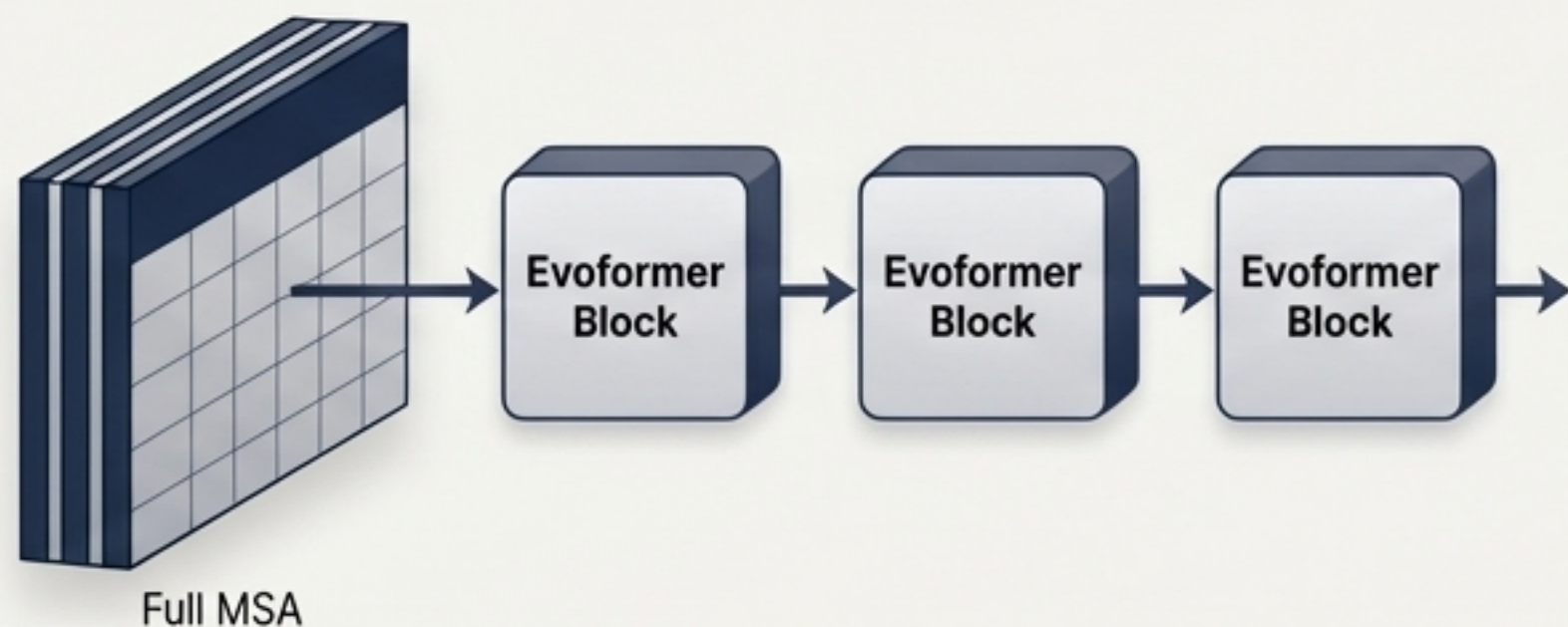
- **Proteins:** 1 Token = 1 Amino Acid Residue
- **Nucleic Acids:** 1 Token = 1 Nucleotide
- **Small Molecules & Covalent Modifications:** 1 Token = 1 Atom

This flexible 'atomic' representation for non-polymeric molecules allows the model to handle any chemical entity within the same framework, creating a universal input system.

The Pairformer: Distilling Co-evolutionary Clues More Efficiently

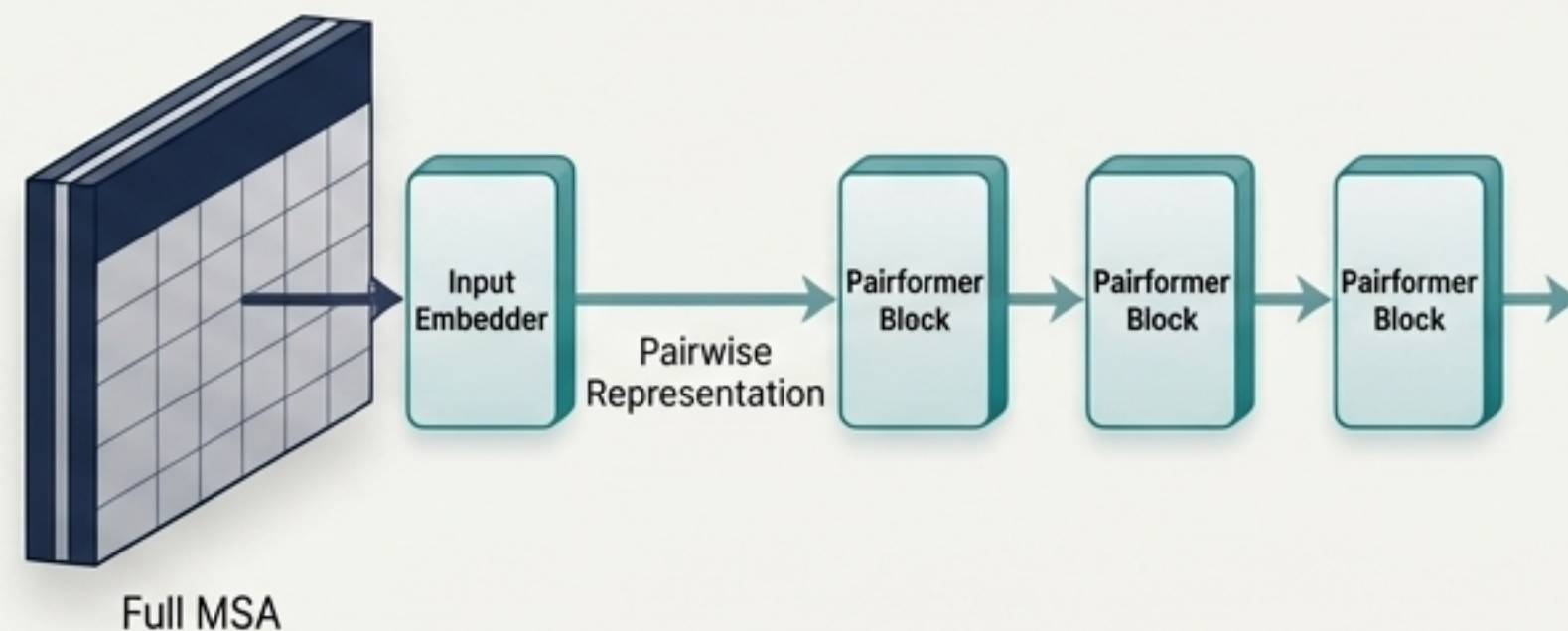
AlphaFold2's Evoformer

Carried the full, bulky MSA representation (sequences $\times$ residues $\times$ channels) throughout its processing blocks.



AlphaFold3's Approach

Processes MSA and template information upfront, "baking" the evolutionary insights into a much more compact pairwise representation early on.



This new approach significantly **reduces memory usage** by eliminating the need to hold the entire MSA representation in memory throughout the main processing loop.

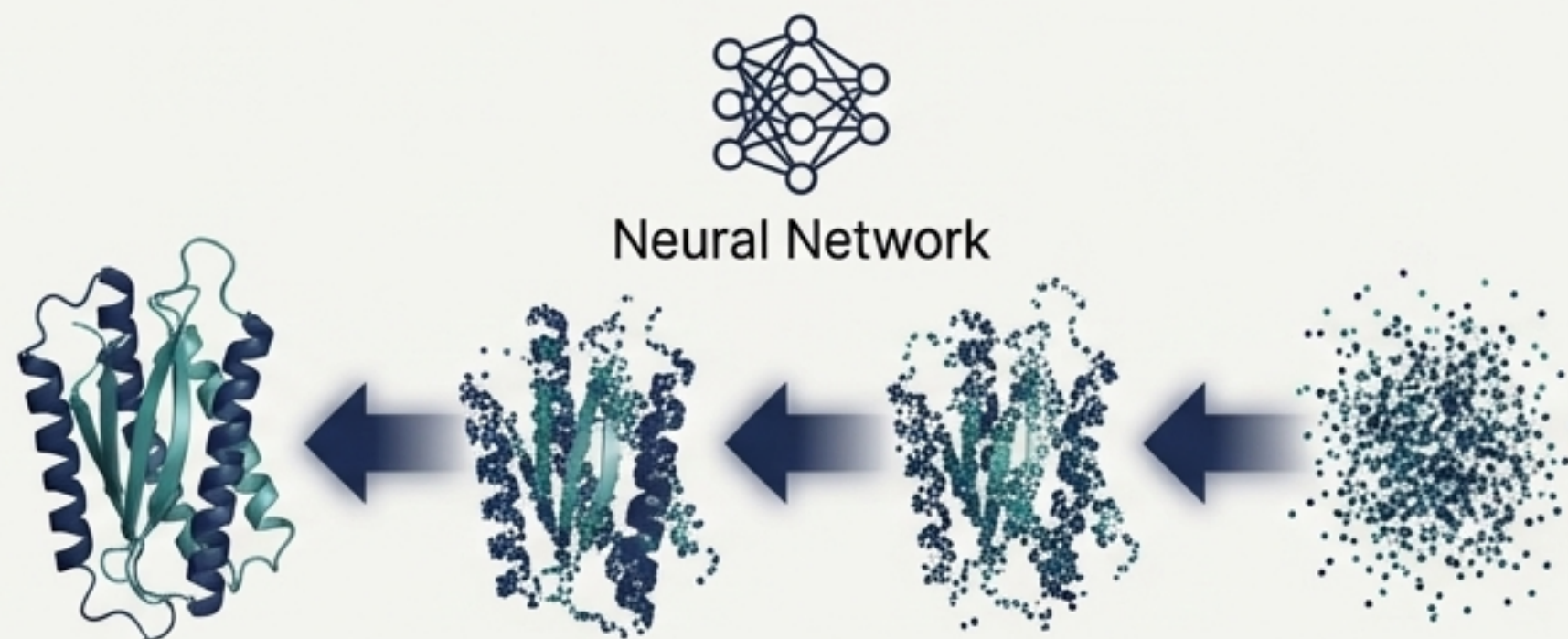
The Core Generative Engine: Building Structures with Diffusion

Training (Forward Process)



Training: Systematically corrupt known structures with noise.

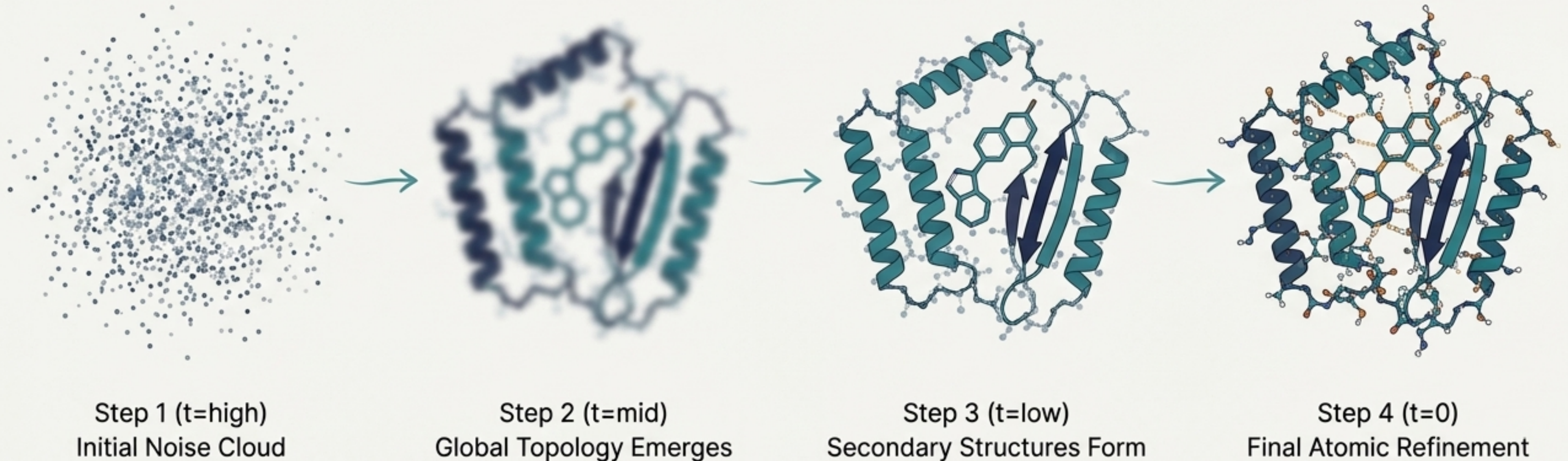
Generation (Reverse Process)



Inference: Train a model to reverse the process, generating a structure from pure noise.

Instead of directly calculating atomic coordinates, AlphaFold3 learns to 'denoise' a random cloud of atoms into a physically plausible structure. This is a stochastic, not deterministic, process.

From Noise to Structure: An Iterative Refinement



The model first establishes the global fold and then spends subsequent steps refining fine-grained atomic details. Because the process is stochastic, each run from the same noise input will produce a slightly different result.

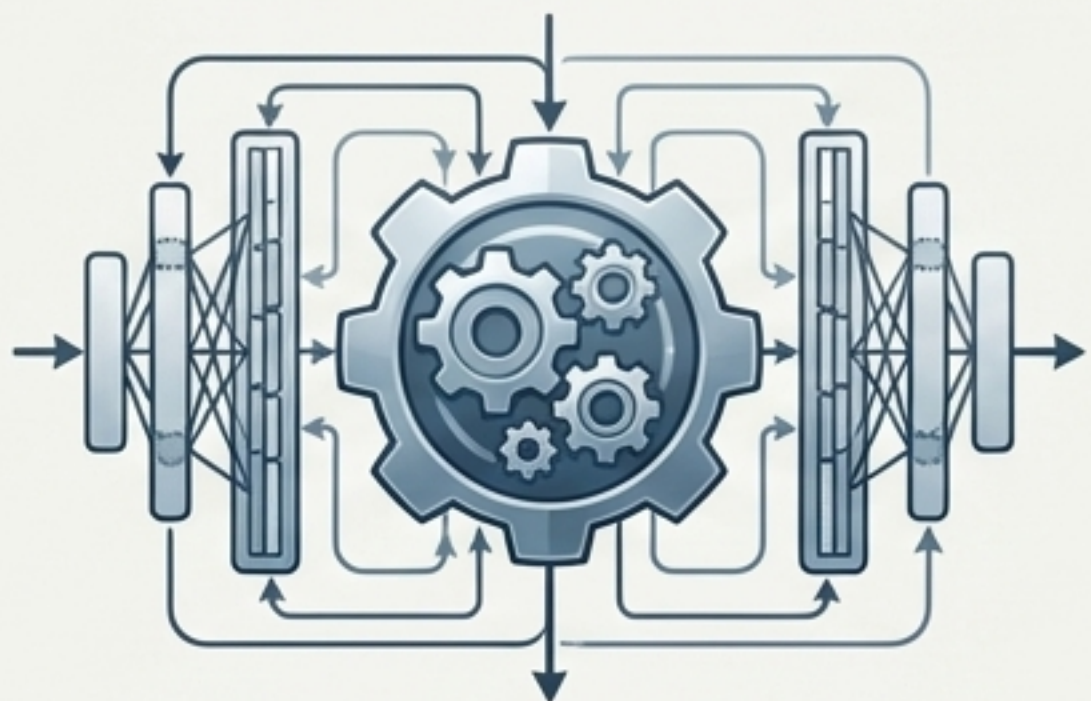
A Surprising Finding: Equivariance Can Be Learned, Not Engineered

The Challenge

Predictions must be invariant to rotation and translation in 3D space (SE(3) equivariance).

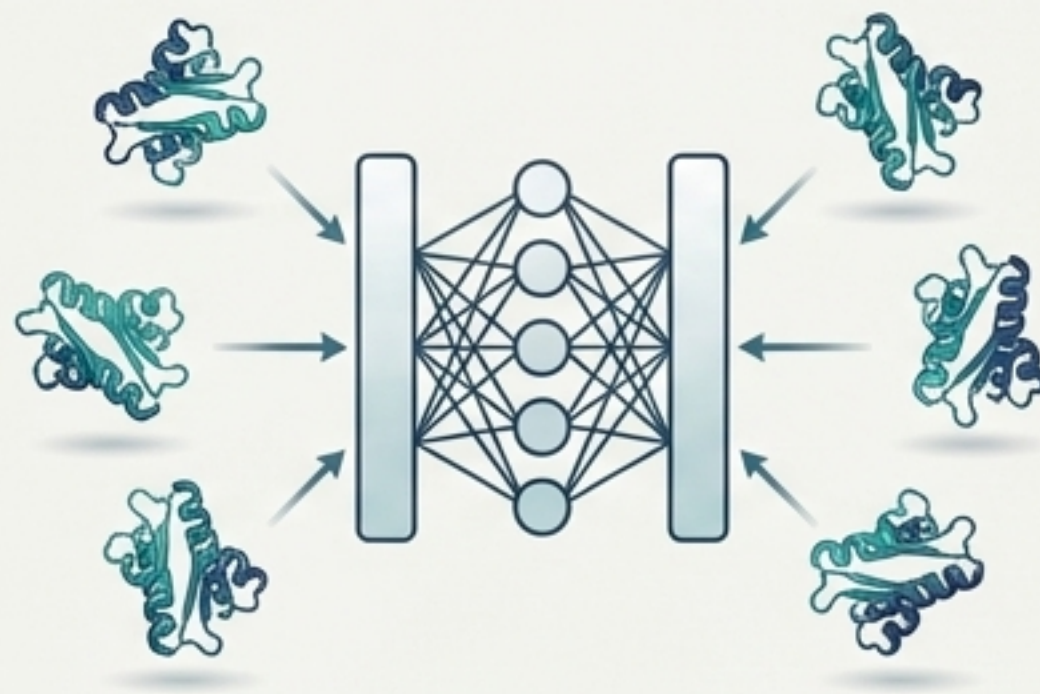
Traditional Approach (e.g., AF2's IPA)

Build geometric constraints directly into the network architecture to enforce this property.



AlphaFold3's Approach

Use standard neural network components but train the model on randomly rotated and translated copies of every structure in the dataset.

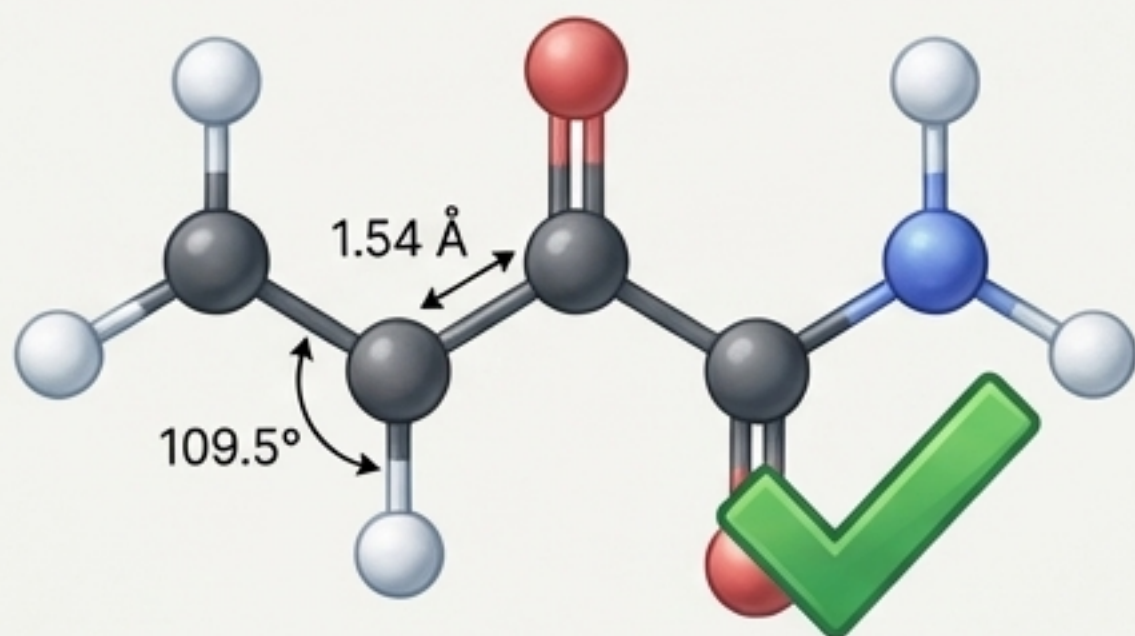


AlphaFold3 demonstrates that a model can learn the principles of 3D geometry from data augmentation alone, sparking a debate on the necessity of explicitly equivariant architectures.

The Double-Edged Sword of Unconstrained Diffusion

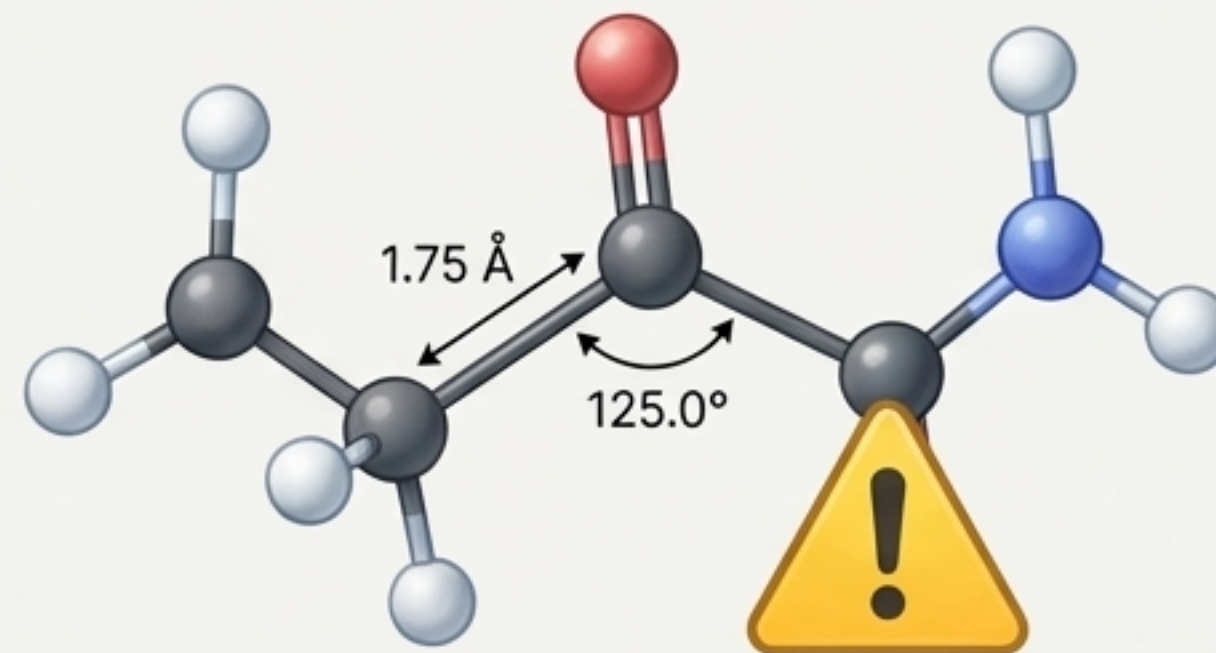
AlphaFold2's FAPE Loss

Explicitly penalized violations of ideal bond lengths, angles, and chirality, acting as a built-in physics engine.



AlphaFold3's Diffusion Loss

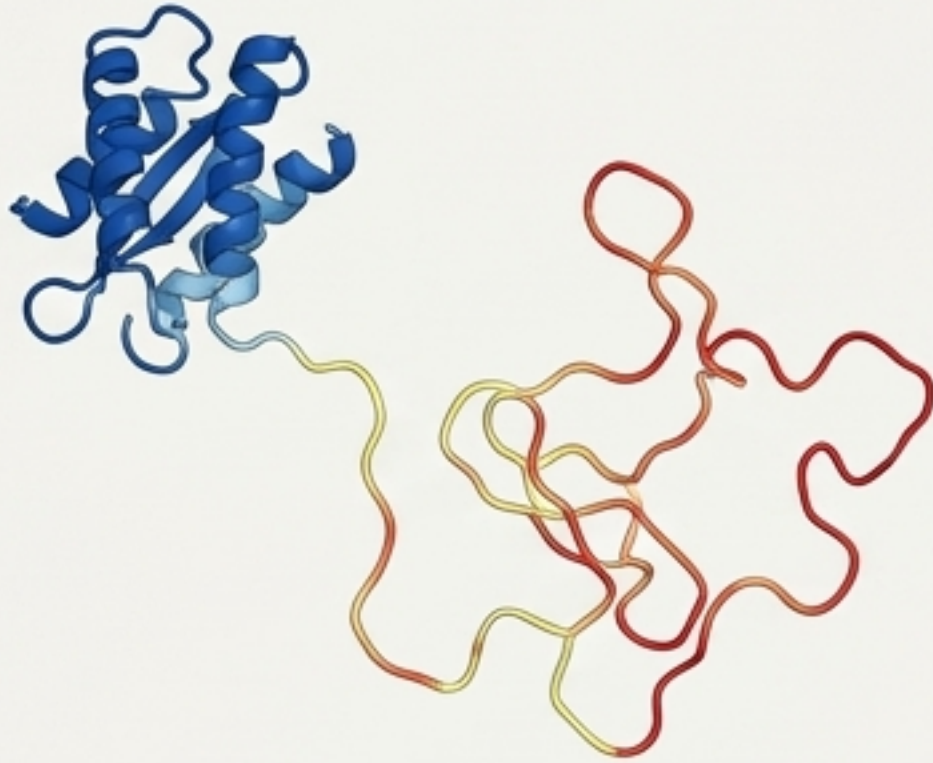
Aims only to match the denoising step, without inherent knowledge of stereochemistry or bond geometry.



The model has the freedom to violate ideal bond geometry and even produce incorrect chirality. This flexibility allows it to model diverse chemistry but requires greater user vigilance and downstream validation.

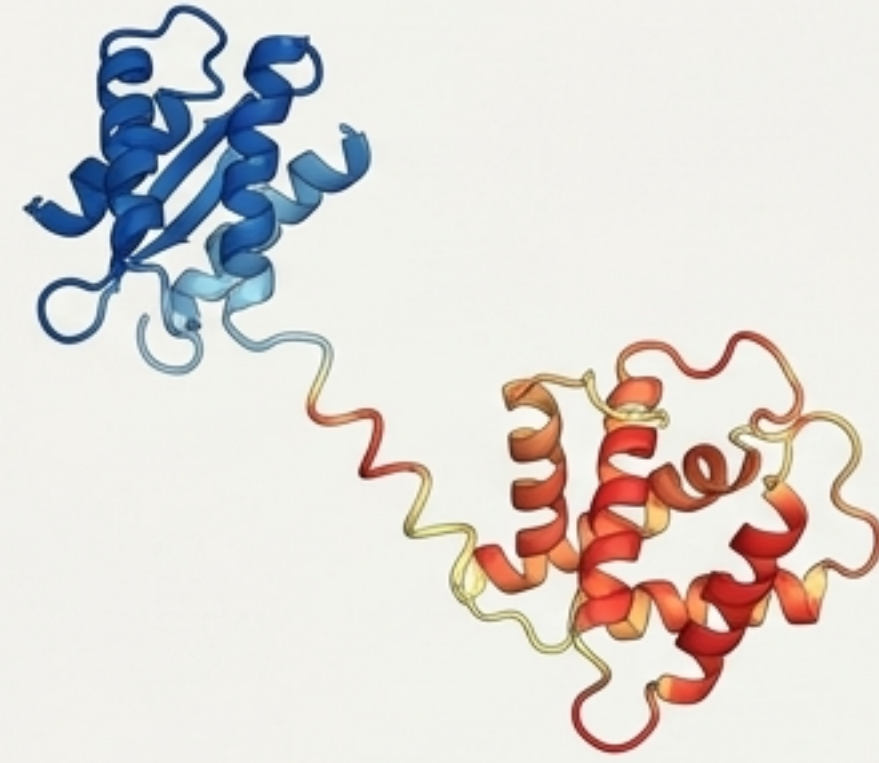
The Peril of Hallucination in Intrinsically Disordered Regions

AlphaFold2 Output



AF2 clearly indicates uncertainty.

AlphaFold3 Output

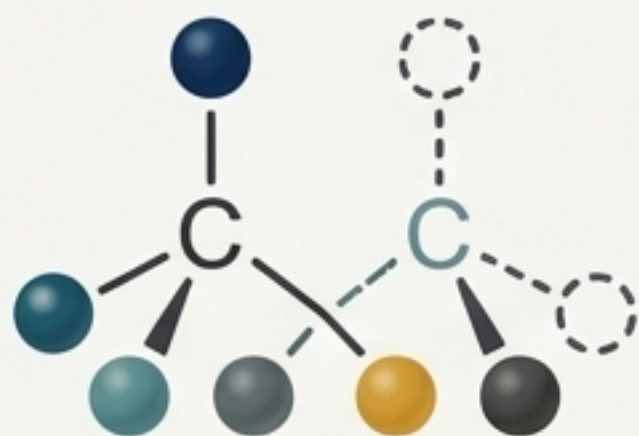


AF3 "hallucinates" structure but correctly reports low confidence.

Diffusion models are trained to produce data-like samples, so AF3 can invent structure where none exists. Always check the pLDDT confidence scores—the model often knows it is hallucinating.

Critical Areas for User Scrutiny

Chirality Errors



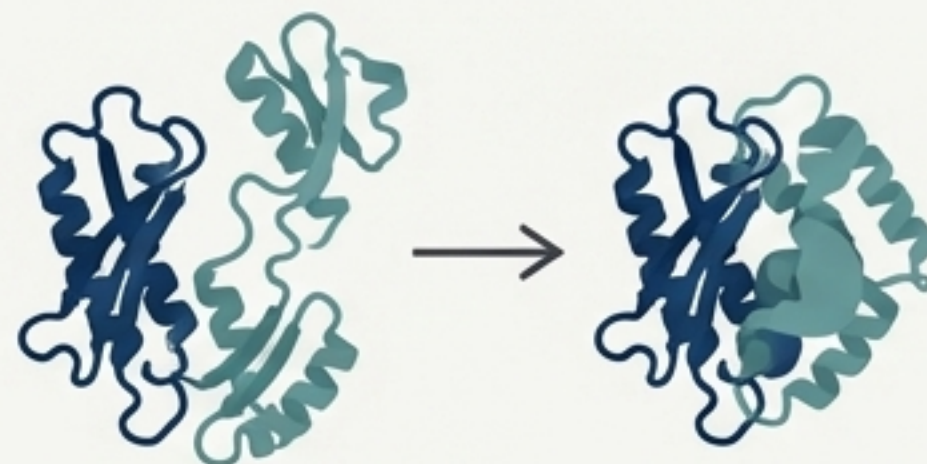
Up to 4-5% of small molecules in the PoseBusters dataset were predicted with incorrect stereochemistry. Manual checking of chiral centers is essential.

Atomic Clashes



The lack of a physics-based loss can lead to sterically impossible structures, including overlapping atoms or entire chains passing through one another.

State Sampling



The model often fails to capture distinct conformational states (e.g., apo vs. holo). It may collapse to a single, dominant conformation seen in the training data, regardless of input context.

The Race to Replicate and Extend

Current Availability

AlphaFold3 is currently accessible only via the public AlphaFold Server for non-commercial research.

Promised Release

A full, open-source release of the code and weights is anticipated.

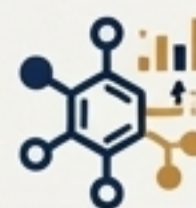
Community Efforts

In the absence of official code, several open-source replication efforts are underway.



HelixFold-3

Replication effort from ByteDance.



KAI-AN

Replication that includes features for incorporating experimental data.



LucidRains' open-source implementation

A large, community-driven effort to reproduce AlphaFold3 in PyTorch.

While the official model remains closed-source for now, the scientific community is actively working to reproduce, validate, and build upon the methods described in the paper.

A New Paradigm Demands a New Set of Rules

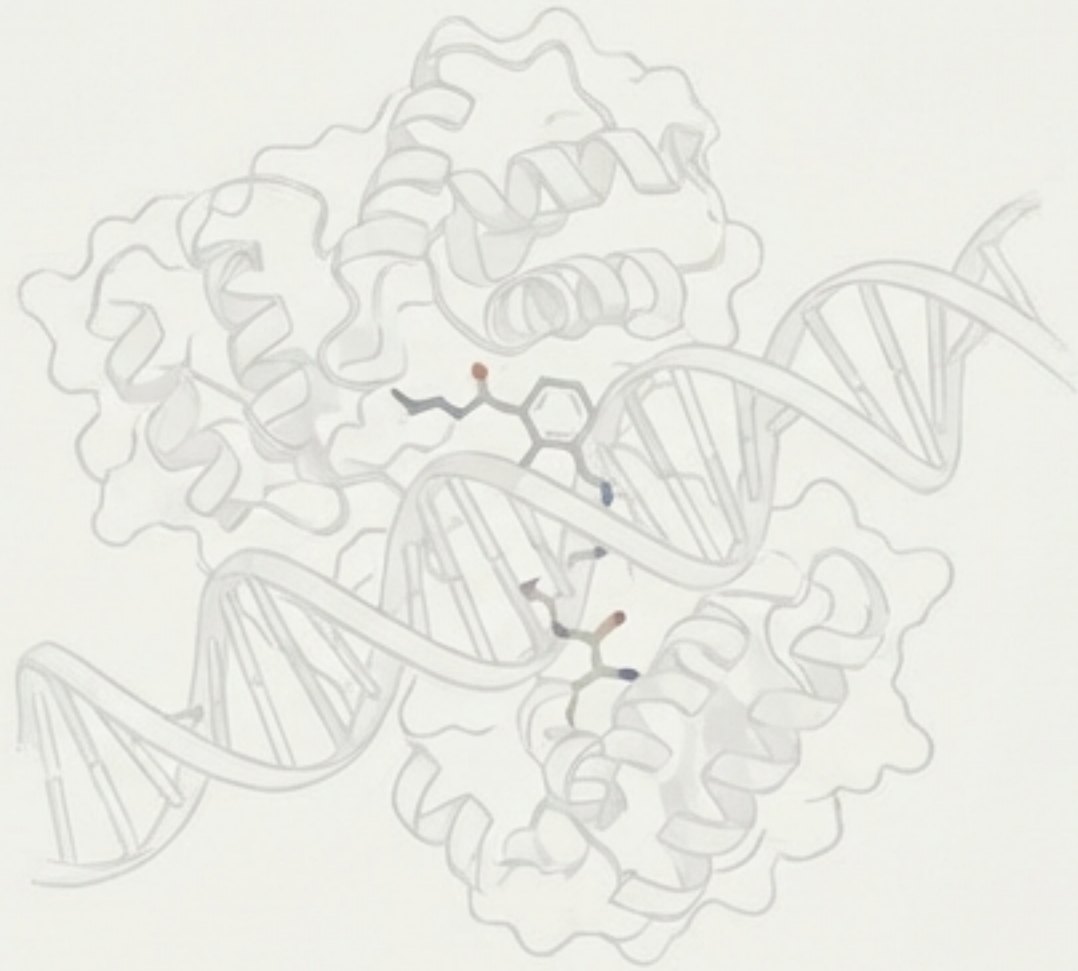
The Power

- **Unified Modeling:** A single model for proteins, nucleic acids, ligands, ions, and modifications.
- **Generative Capability:** Diffusion enables the creation of novel structures from noise.
- **Architectural Efficiency:** The Pairformer reduces memory load by distilling MSA information upfront.

The Responsibility

- **Sample Aggressively:** Hard targets require thousands of predictions to find correct conformations.
- **Validate Geometry:** Manually check for incorrect chirality and steric clashes.
- **Trust, But Verify:** Heed confidence scores to avoid being misled by plausible-looking “hallucinations” in disordered regions.

Q&A



Source Material & Further Reading:

DeepMind & Isomorphic Labs, 'Accurate structure prediction of biomolecular interactions with AlphaFold 3', Nature (2024)
RosettaCommons, 'Structure Prediction with AlphaFold3', YouTube (2024)